AI Label Reader for Accessibility

Requirements Document | Version 0.1.0 | Software Construction and Development

1. Project title and group members

Project title: AI Label Reader for Accessibility

Name	Roll number	GitHub username	Initial responsibility
Roshaan rehman	38	roshaanrehman	Requirements documentation and OCR module
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2. Problem description

People with low vision or reading difficulties (such as dyslexia) often struggle to read the small, dense text on everyday products: medicine boxes, food packaging, cleaning products and printed notices. Missing a dosage instruction, an allergen warning or an expiry date can be inconvenient or unsafe. Existing solutions are either expensive, require internet-heavy setups, or read text without making it easier to understand.
This project builds a simple application that lets a user photograph a label, extracts the text using OCR, reads it aloud, and offers a short plain-language summary that links back to the original text.

3. Objectives and intended users

Objectives

- Allow a user to extract readable text from a photograph of a product label.
- Read the extracted text aloud using text-to-speech.
- Provide a simple-language summary of the label while keeping the original text accessible.
- Reuse existing OCR and text-to-speech tools instead of building custom AI models.

- Practise collaborative development using Git and an agile workflow.

Intended users

- People with low vision.
- People with reading difficulties such as dyslexia.
- Elderly users who find small print hard to read.
- Caregivers helping others read labels.

4. Initial functional requirements

ID	Requirement
FR-1	The system shall allow the user to upload or capture an image of a product label.
FR-2	The system shall extract text from the image using an OCR library or API.
FR-3	The system shall display the extracted text on screen in a clear, readable format.
FR-4	The system shall read the extracted text aloud using text-to-speech.
FR-5	The user shall be able to pause, resume and stop the audio.
FR-6	The system shall generate a short, simple-language summary of the extracted text.
FR-7	The summary shall include a link or reference back to the full original text.
FR-8	The system shall show a clear message if no text can be detected or the image quality is too low.
FR-9	The user shall be able to retry with a new image.

5. Non-functional requirements

- **Accessibility:** large fonts, high-contrast display, simple layout and screen-reader friendly controls.
- **Usability:** a new user should be able to scan a label and hear it read within a few steps and without training.
- **Performance:** text extraction and audio should start within a reasonable time (target: under 10 seconds on a typical image).
- **Accuracy:** OCR should handle clear, well-lit labels; the system should warn the user when confidence is low.
- **Reliability:** the app should not crash on blurry, empty or unsupported images.
- **Privacy:** images should be processed only for text extraction and not stored permanently without the user's knowledge.
- **Maintainability:** modular code (OCR, text-to-speech and summarization as separate modules) with clear naming.

6. Prioritized feature list (product backlog)

Priority	Feature	Related requirement
High	Image upload or capture	FR-1
High	OCR text extraction	FR-2, FR-3
High	Text-to-speech playback	FR-4
High	Error handling for unreadable images	FR-8, FR-9
Medium	Audio controls (pause, resume, stop)	FR-5
Medium	Simple-language summary	FR-6, FR-7
Medium	Accessible, high-contrast interface	NFR
Low	Adjustable speech speed and font size	Enhancement
Low	Multiple language support	Enhancement

Priority	Feature	Related requirement
Low	History of recently scanned labels	Enhancement

This backlog will be maintained in the repository (GitHub Issues or Projects board) and updated as the project evolves.

7. Assumptions, constraints and limitations

Assumptions

- Users can take or provide a reasonably clear photograph of the label.
- Labels are printed in a language supported by the chosen OCR tool (initially English).
- Users have a device with a camera or image files, and audio output.

Constraints

- The project is developed in Python using existing OCR and text-to-speech libraries or APIs.
- Development time is limited to the semester schedule.
- Free tools or free API tiers will be preferred.

Known limitations

- OCR accuracy drops with blur, glare, curved packaging, decorative fonts and very small print.
- The summary is produced by AI and may miss details, so users are directed back to the original text, especially for medicines and allergens.
- The app is a reading aid and not a substitute for professional medical or safety advice.

8. Note on evolving requirements

In line with an agile approach, these requirements are an initial baseline. They will be reviewed and updated as the team learns more, and every change will be traceable through Git commits and version tags (starting at v0.1.0).